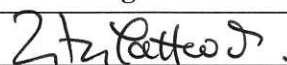
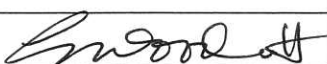
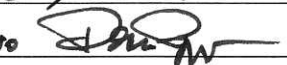
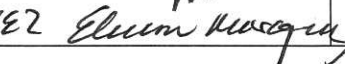


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| <p>Female Sprague-Dawley rats were rendered diabetic by a single IP injection of STZ and hyperglycemia was confirmed four days after injection. Starting two weeks after the onset of diabetes, but prior to the manifestation of tactile allodynia (i.e. disease prevention mode), animals were dosed both intravenously and intrathecally to ensure 504B3 access to both the peripheral and central nervous system compartments, as the literature is unclear as to which or both compartments exhibit dysregulated LPA levels. Animals were then tested weekly for the presence of hyperalgesia and tactile allodynia. Results showed that anti-LPA antibody treatment increased the paw withdrawal threshold in diabetic rats to a level similar to that observed in non-diabetic rats, indicating a marked reduction in mechanical allodynia. In summary, this study demonstrated efficacy of the murine anti-LPA antibody to prevent mechanical allodynia in the STZ rat model of type 1 diabetes.</p> |  |                                                               |                                         |                                                               |                                                                             |                                                        |                                                              |  |

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| <b>Title: Efficacy of murine anti-LPA Ab in the STZ rat model of type 1 diabetes – Preliminary Study</b> | Page 1 of 15 |

**Efficacy of murine anti-LPA Ab in the STZ rat model of type 1 diabetes - Preliminary Study**

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## **1 SUMMARY**

The purpose of this study was to test the efficacy of the murine anti-lysophosphatidic acid (LPA) antibody B3 to ameliorate pain behavior in the STZ rat model of type 1 diabetes.

Female Sprague-Dawley rats were rendered diabetic by a single IP injection of STZ and hyperglycemia was confirmed four days after injection. Starting two weeks after the onset of diabetes, but prior to the manifestation of tactile allodynia (i.e. disease prevention mode), animals were dosed both intravenously and intrathecally to ensure B3 access to both the peripheral and central nervous system compartments, as the literature is unclear as to which or both compartments exhibit dysregulated LPA levels. Animals were then tested weekly for the presence of hyperalgesia and tactile allodynia. Results showed that anti-LPA antibody treatment increased the paw withdrawal threshold in diabetic rats to a level similar to that observed in non-diabetic rats, indicating a marked reduction in mechanical allodynia.

In summary, this study demonstrated efficacy of the murine anti-LPA antibody to prevent mechanical allodynia in the STZ rat model of type 1 diabetes.

## **2 INTRODUCTION**

The bioactive lipid lysophosphatidic acid (LPA) is a key extracellular signaling molecule in neuropathic pain. LPA and its receptors are responsible for long-lasting mechanical allodynia and thermal hyperalgesia as well as demyelination and upregulation of pain-related protein expression patterns in a variety of animal models. Evidence supports the hypothesis that antagonism of the LPA signaling pathways would benefit the treatment of neuropathic pain. We have developed the first monoclonal antibody targeting LPA (anti-LPA Ab). In preliminary studies, Lpath's anti-LPA mAb has demonstrated efficacy in the Seltzer model of neuropathic pain where the literature has demonstrated a role for LPA in the disorder, we next turned to diabetic pain models where LPA has not heretofore been implicated. Diabetic neuropathy in type 1 and 2 diabetes in both humans and animal models is characterized by pathophysiological changes in all components (sensory, motor and autonomic) of the peripheral nervous system. In addition to the changes of primary afferent nerves, central sensitization is believed to be an important mechanism underlying persistent pain, including neuropathic and inflammatory pain. The efficacy of our anti-LPA mAbs in the Seltzer model, in part, validates LPA as a target for nerve-injury induced neuropathic pain. Thus, if LPA plays a role in diabetes-induced neuropathic pain as it does in the Seltzer model, we postulated that treatment of diabetic rats with a murine anti-LPA antibody (504B3) would ameliorate pain behaviors associated with diabetes. In order to explore this possibility, studies were performed in the laboratory of Dr. Nigel Calcutt at UCSD using the well-established rat model of STZ-induced type 1 (insulin deficient) diabetes. The STZ model uses tactile allodynia as a behavioral index of diabetes-induced neuropathic pain.

### **3 MATERIALS AND METHODS**

#### **3.1 Test Article and Control**

Murine anti-LPA antibody (B3) Lot# HR1172

#### **3.2 STZ Model Development**

Female Sprague-Dawley rats (Harlan Industries, San Diego CA) weighing 225-250 grams each were maintained at room temperature, between 65 to 82°F with relative humidity between 30 to 70%. The room was illuminated with fluorescent lighting on a daily 12 hour light/dark cycle. All animals were maintained 2/cage with free access to dry food and municipal water. Insulin deficient diabetes was induced following an overnight fast by a single IP injection of STZ (55 mg/kg) dissolved in 0.9% sterile saline. Hyperglycemia was confirmed 4 days later and also prior to behavioral testing in a sample of blood obtained by tail prick using a strip operated reflectance meter. All animals were observed daily and weighed regularly during the study period.

#### **3.3 Sample Collection and Analysis**

Blood samples were collected in tubes containing potassium EDTA (K<sub>2</sub>EDTA) predose (prior to dosing), and 24 hours after the last dose each week for the duration of the study. Blood samples were centrifuged at approximately 5000 rpm for approximately 10 minutes at 4°C, and plasma was harvested. Plasma samples were immediately frozen in dry ice and stored at -70 ± 10°C until assayed. Bioactive LPA was estimated from the ability of LPA present in plasma samples to activate the LPA<sub>1</sub> receptor in the DiscoverX reporter cell assay.

Bioactive LPA was detected using The CHO-K1 EDG2 (LPA<sub>1</sub> receptor)  $\beta$ -arrestin clone (PathHunter™) that uses Enzyme Fragment Complementation technology in which two weakly complementing fragments of the B-galactosidase ( $\beta$ -gal) enzyme are expressed in stably transfected cells. One fragment of the  $\beta$ -gal, termed the enzyme acceptor (EA), is fused to the C-terminus of  $\beta$ -arrestin2. The complementing fragment of  $\beta$ -gal, termed the ProLink™ tag, is expressed as a C-terminal fusion protein with the LPA<sub>1</sub> receptor. Upon activation, the LPA<sub>1</sub> receptor is phosphorylated, providing a binding site for  $\beta$ -arrestin. The interaction of  $\beta$ -arrestin and the LPA<sub>1</sub> receptor forces the interaction of ProLink and EA, thus allowing complementation of the two fragments of  $\beta$ -gal and the formation of functional enzyme capable of hydrolyzing substrate and generating a chemiluminescent signal. Briefly, CHO-K1 EDG2 cells were plated at and incubated at 37°C/5% CO<sub>2</sub>. After 24 hours, plates were starved with reduced serum media at 37°C for another 24 hours. LPA standard curves were prepared by titrating LPA (0 – 60  $\mu$ M) in F12 media containing 1 mg/mL BSA onto the plate to determine low and maximal signal (dotted lines on [Figure 2](#)). Standards and samples were added to the plate and incubated at room temperature for 90 minutes. The plates were

washed, incubated with the detection reagent in the dark for 90 minutes and read on a standard luminescence plate reader.

### 3.4 Tactile Response Threshold

Rats were transferred to a testing cage with a wire mesh bottom and allowed to acclimate. Von Frey filaments (Stoelting, Wood Dale IL) were used to determine the 50% mechanical threshold for foot withdrawal. A series of filaments, starting with one possessing a buckling weight of 2.0g, were applied in sequence to the plantar surface of the right hindpaw with a pressure that causes the filament to buckle. Lifting of the paw was recorded as a positive response and the next lightest filament chosen for the next measurement. Absence of a response after 5 seconds prompted use of the next filament of increasing weight. This paradigm was continued until four measurements were made after an initial change in the behavior or until five consecutive negative (given the score of 15g) or four positive (score of 0.25g) scores occurred. The resulting sequence of positive and negative scores was used to interpolate the 50% response threshold. Only rats with a 50% tactile response threshold below 6 g are considered allodynic and brought forward for drug testing.

### 3.5 Study Design

**Prevention study:** To investigate the efficacy of B3 in preventing the development of diabetic induced neuropathic pain, three groups of diabetic rats were established by treatment with STZ. Starting two weeks after the onset of diabetes, but prior to the manifestation of tactile allodynia, animals were dosed both intravenously and intrathecally to ensure B3 access to both the peripheral and central nervous system compartments, as the literature is unclear as to which or both compartments exhibit dysregulated LPA levels. The dosing regimens for the treatment groups were: two groups of test animals received 10 mg/kg of intravenously administered B3 twice a week. In addition to the intravenously administered antibody, animals received B3 twice weekly via an intrathecal catheter. One group of animals (“Low dose +D” in [Figure 1](#)) received 2 µg B3 and one group of animals (“high dose+D” in [Figure 1](#)) received 10 µg B3 intrathecally. One group of diabetic animals received vehicle alone (“Veh+D” in [Figure 1](#)). Nondiabetic rats were used as controls (Veh+C in [Figure 1](#)). The tactile response threshold was measured at the start of week 3, 4 days after the last treatment.

**Intervention study:** Based on encouraging results in the prevention study ([Figure 2](#)), the nine control diabetic rats, Veh + C group from it, were then used to assess the reversibility of the already-established diabetic-induced neuropathic pain and to discriminate between intravenous and intrathecal administration of B3. The rats received the following treatments on different days: gabapentin 50mg/Kg (intraperitoneal) as a control agent on day 1, B3 i.t. (10 ug/10 µL) on day 3, B3 i.v. (10 mg/Kg) on day 6 and the combination (i.t. 10 ug/10 µL + i.v. 10 mg/Kg) on day 12. After each dosing, subsequent measurements of tactile allodynia (PWT response) at 1, 3 and 6 hr were recorded as described above for the prevention study.

### 3.6 Data Analyses

All data was generated and collected at the laboratory of Dr. Nigel Calcutt at UCSD. Descriptive statistics was used to calculate median, mean, standard deviation (SD) and standard error of the mean (SEM) using Microsoft Excel. An excel spread sheet containing all data and analysis was sent via e-mail to Lpath. GraphPad Prism software (version 6.0, Graph Software inc, La Jolla, CA) was used to graph and analyze (ANOVA) the data.

**Table 1 Prevention Study**

Dose groups

|                       |                     |  |
|-----------------------|---------------------|--|
| Group 1 "Veh+C" *     |                     |  |
| Antibody/Condition    | Control nondiabetic |  |
| Lot Number            | N/A                 |  |
| Dose level            |                     |  |
| Dose Route            |                     |  |
| Frequency             |                     |  |
| Total Doses           |                     |  |
| Group 2 "Veh+D"       |                     |  |
| Antibody/Condition    | STZ + Vehicle       |  |
| Lot Number            | N/A                 |  |
| Dose level            |                     |  |
| Dose Route            | it and iv           |  |
| Frequency             | 2x week             |  |
| Total Doses           | 4it and 4 iv        |  |
| Group 3 "Low dose +D" |                     |  |
| Antibody/Condition    | STZ + B3            |  |
| Lot Number            | HR1172              |  |
| Dose level            | 2ug/it + 10mg/Kg    |  |
| Dose Route            | it and iv           |  |
| Frequency             | 2x week             |  |
| Total Doses           | 4it and 4 iv        |  |
| Group 4 "high dose+D" |                     |  |
| Antibody/Condition    | STZ + B3            |  |
| Lot Number            | HR1172              |  |
| Dose level            | 10ug/it + 10mg/Kg   |  |
| Dose Route            | it and iv           |  |
| Frequency             | 2x week             |  |
| Total Doses           | 4it and 4 iv        |  |

\* Control group was treated with gabapentin as positive control and after washout used for study 2

**Table 2 Intervention Study, Dose Schedule. The Control Diabetic rats from the prevention study (n=9) were used for this study**

|       |                                                                               |  |  |
|-------|-------------------------------------------------------------------------------|--|--|
|       | Treatment: Diabetic rats were injected with Gabapentin 50 mg/kg (2 ml/Kg, ip) |  |  |
| Day 0 | Gabapentin was dissolved in saline solution                                   |  |  |
|       | Tactile Allodynia was measured every hour                                     |  |  |

|        |                                                                           |  |  |
|--------|---------------------------------------------------------------------------|--|--|
|        | Treatment: Diabetic rats were injected with Anti-LPA mAb 10 ug/10 ul (it) |  |  |
| Day 21 | Anti-LPA mAb was diluted in PBS as vehicle                                |  |  |
|        | Tactile Allodynia was measured every hour                                 |  |  |

|        |                                                                         |  |  |
|--------|-------------------------------------------------------------------------|--|--|
|        | Treatment: Diabetic rats were injected with Anti-LPA mAb 10 mg/Kg (i.v) |  |  |
| Day 23 | Anti-LPA mAb was diluted in PBS as vehicle                              |  |  |
|        | Tactile Allodynia was measured every hour                               |  |  |

|        |                                                                                              |  |  |
|--------|----------------------------------------------------------------------------------------------|--|--|
|        | Treatment: Diabetic rats were injected with Anti-LPA mAb 10 ug/10 ul (l.t.) + 10 mg/kg (i.v) |  |  |
| Day 28 | Anti-LPA mAb was diluted in PBS as vehicle                                                   |  |  |
|        | Tactile Allodynia was measured every hour                                                    |  |  |

## 4 RESULTS

### 4.1 Body Weight

Diabetic rats had a lower weight than control rats by the end of the study. The mean ( $\pm$ SEM) body weight for diabetic rats at the conclusion of the study ( $201 \pm 4$  g) was significantly lower ( $P < 0.05$  by one-way ANOVA with Bonferroni *post-hoc* analysis) than the corresponding mean ( $\pm$ SEM) body weight of the control group ( $223 \pm 3$  g). There was no significant difference in weight between, vehicle and 504B3 antibody treated rats (Table 3).

Table 3 Terminal Body Weight and Blood Glucose

| LPA Study                           |                 |            | 12-Jul-10                            |                 |            |
|-------------------------------------|-----------------|------------|--------------------------------------|-----------------|------------|
| C+Veh                               |                 |            | D+Veh                                |                 |            |
| Rat #                               | Glucose (mg/dl) | Weight (g) | Rat #                                | Glucose (mg/dl) | Weight (g) |
| 1.0                                 | 115.0           | 211.0      | 5.0                                  | Hi              | 205.4      |
| 1.1                                 | 98.0            | 214.0      | 5.1                                  | 494.0           | 186.0      |
| 1.2                                 | 154.0           | 216.8      | 6.0                                  | Hi              | 207.7      |
| 2.0                                 | 112.0           | 224.1      | 6.1                                  | 502.0           | 202.0      |
| 2.1                                 | 101.0           | 220.0      | 7.1                                  | Hi              | 212.2      |
| 3.0                                 | 110.0           | 234.6      | 12.0                                 | 580.0           | 181.7      |
| 3.1                                 | 91.0            | 230.5      | 13.0                                 | 436.0           | 212.0      |
| 3.2                                 | 86.0            | 228.9      | 13.1                                 | Hi              | 200.6      |
| 4.0                                 | 97.0            | 231.6      | 14.1                                 | 58.0            | 133.0      |
| 4.1                                 | 88.0            | 225.1      |                                      |                 |            |
| Mean                                | 105.2           | 223.7      | Mean                                 | 503.0           | 201.0      |
| Median                              | 99.5            | 224.6      | Median                               | 498.0           | 203.7      |
| SEM                                 | 6.6             | 2.7        | SEM                                  | 20.9            | 4.0        |
| D+Anti-LPA mAb 2 ug it; 10 mg/kg iv |                 |            | D+Anti-LPA mAb 10 ug it; 10 mg/kg iv |                 |            |
| Rat #                               | Glucose (mg/dl) | Weight (g) | Rat #                                | Glucose (mg/dl) | Weight (g) |
| 8.0                                 | Hi              | 176.0      | 7.0                                  | 504.0           | 212.4      |
| 9.0                                 | Hi              | 170.0      | 10.1                                 | Hi              | 160.0      |
| 9.1                                 | 577.0           | 171.3      | 11.0                                 | 551.0           | 178.0      |
| 10.0                                | Hi              | 191.1      | 12.1                                 | Hi              | 181.0      |
| 11.1                                | 553.0           | 165.1      | 17.0                                 | Hi              | 175.7      |
| 14.0                                | 548.0           | 190.0      | 17.1                                 | 473.0           | 194.5      |
| 15.0                                | Hi              | 171.1      | 18.0                                 | 566.0           | 160.6      |
| 15.1                                | 442.0           | 159.0      | 18.1                                 | Hi              | 203.1      |
| 16.0                                | 521.0           | 233.8      | 19.0                                 | 450.0           | 157.0      |
| 16.1                                | 389.0           | 212.0      | 19.1                                 | 579.0           | 168.0      |
| Mean                                | 505.0           | 183.9      | Mean                                 | 520.5           | 179.0      |
| Median                              | 534.5           | 173.7      | Median                               | 527.5           | 176.9      |
| SEM                                 | 24.5            | 7.8        | SEM                                  | 17.6            | 6.3        |
| Hi ≥ 600 or limit of detection      |                 |            |                                      |                 |            |

## 4.2 Blood Glucose

Blood glucose concentration was increased approximately 5-fold in diabetic rats compared with control lean group (Table 3). The mean ( $\pm$ SEM) blood glucose for diabetic rats at the end of the study ( $503 \pm 21$  dl/ml) was significantly higher ( $P < 0.05$  by one-way ANOVA with Bonferroni post-hoc analysis) than the corresponding mean ( $\pm$ SEM) blood glucose of the control group ( $105 \pm 6$  dl/ml). There was no significant difference in blood glucose between, vehicle and anti-LPA antibody treated diabetic rats (Table 3).

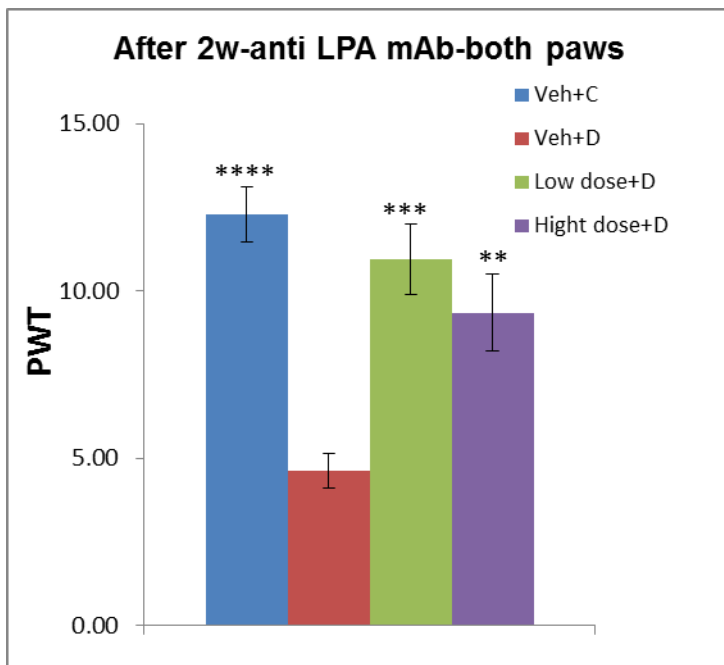
### 4.3 Mechanical Allodynia

**Prevention study:** The results are shown in [Figure 1](#). B3 increased the paw withdrawal threshold in diabetic rats to a level similar to that observed in non-diabetic rats, indicating a marked reduction in allodynia in rats with diabetes-induced neuropathic pain (\*\*\*\*p <0.0001 Vehicle Control v. Vehicle Diabetic, \*\*\*p <0.001 Low Dose v. Vehicle Diabetic and \*\*p <0.01 for High Dose v. Vehicle Diabetic by one-way ANOVA with Bonferroni *post-hoc* analysis).

**Intervention Study:** The results of this preliminary intervention study demonstrate that a single dose of B3 was as effective as gabapentin in ameliorating established diabetic induced neuropathic pain ([Figure 2](#)). Both intravenous antibody administration and intrathecal administration had no significant effect when compared with vehicle treated animals. Although a trend toward efficacy was observed in both groups. Efficacy after intravenous dosing is generally considered to demonstrate involvement of peripheral nervous system mechanisms while a response seen after i.t. dosing is suggestive of central nervous system involvement. The ability of B3 to partially mitigate pain responses from both routes of administration could suggest that LPA mediates both peripheral and central mechanisms of pain transmission. This is supported by the observed significant additive effect when animals were dosed with B3 via both routes of administration in combination (\*\*p <0.01 for anti-LPA i.t. 10 ug/10 µL + i.v. 10 mg/Kg v. Veh by one-way ANOVA with Bonferroni *post-hoc* analysis).

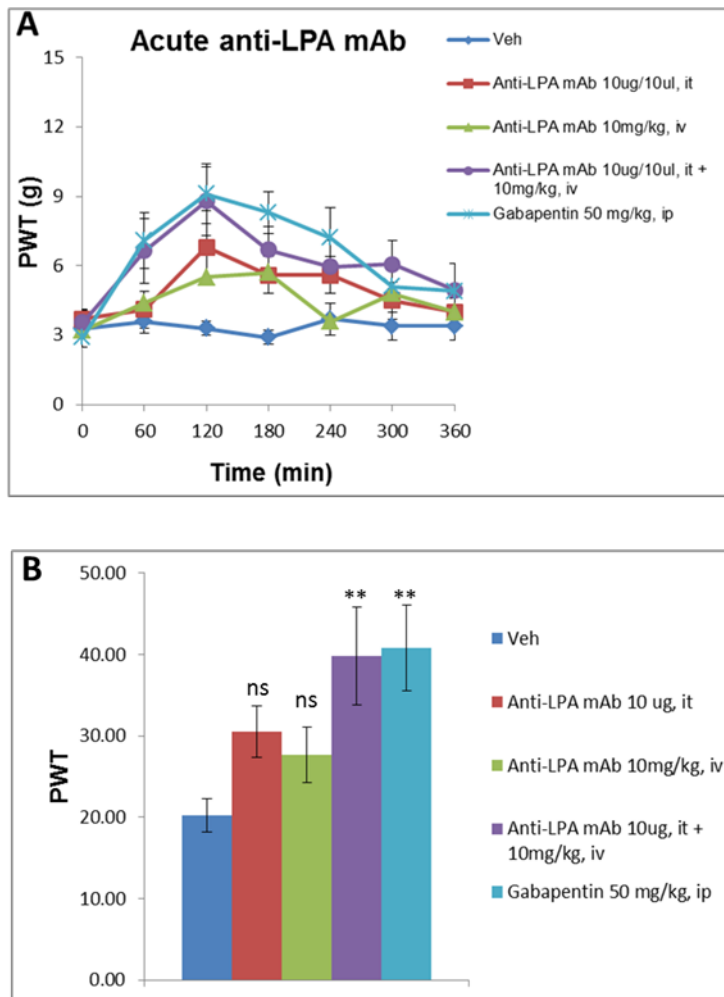
**Bioactive LPA:** The levels of bioactive LPA (defined as the amount of LPA present in plasma to activate the LPA1 receptor in the DiscoverX reporter cell assay) were measured in the plasma of rats receiving intrathecal and intravenous injection of vehicle (Veh), or diabetic (D) rats randomized in the following groups: Veh + D, intrathecal and intravenous injection of vehicle, Low dose + D, intravenous injection anti-LPA-mAb (10 mg/kg) + intrathecal injection of anti-LPA-mAb (2 ug/10 µL) and High dose + D intravenous injection anti-LPA-mAb (10 mg/kg) + intrathecal injection of anti-LPA-mAb (10 ug/10 µL). We demonstrated that there was a significant increase in bioactive LPA in the diabetic animals which was reduced in animals treated with 504B3 antibody ([Figure 3](#)). (\*\*p <0.01 for Veh + D v. Veh and \*p <0.05 for Low dose + D v. Veh by one-way ANOVA with Bonferroni *post-hoc* analysis).

Figure 1. B3 reduces neuropathic pain in diabetic-induced neuropathic pain

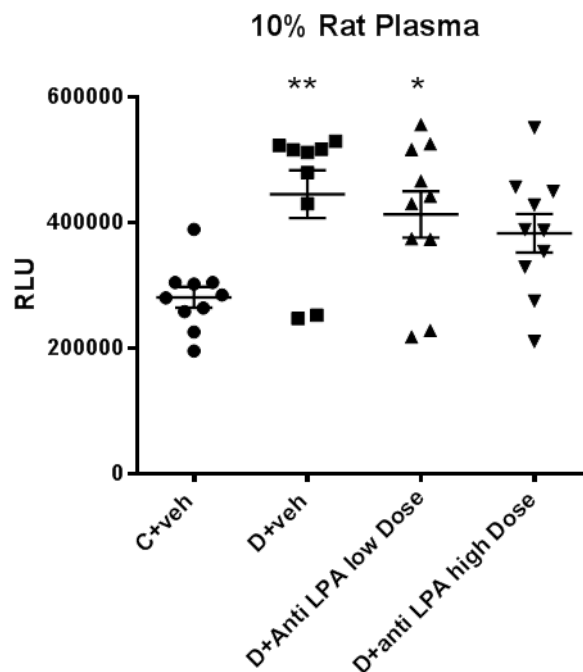


Tactile responses, represented as 50% Paw Withdrawal Threshold (PWT) in control rats, receiving intrathecal and intravenous injection of vehicle (Veh), or diabetic (D) rats randomized in the following groups: Veh + D, intrathecal and intravenous injection of vehicle, Low dose + D, intravenous injection anti-LPA-mAb (10 mg/kg) + intrathecal injection of anti-LPA-mAb (2 ug/10µL) and High dose + D intravenous injection anti-LPA-mAb (10 mg/kg) + intrathecal injection of anti-LPA-mAb (10 ug/10µL). Rats were dosed both, intravenously and intrathecally twice a week and tactile responses were recorded 3 days after last dose. Data are expressed as mean ± SEM, n = 8-9/group. \*\*\*\*p < 0.0001 Veh v. Veh + D, \*\*\*p < 0.001 Low Dose + D v. Veh + D and \*\*p < 0.01 for High Dose v. Veh + D by one-way ANOVA with Bonferroni *post-hoc* analysis.

Figure 2. Effect of B3 on established neuropathic pain.



(A) Temporal Course Curve: Tactile responses, represented as 50% Paw Withdrawal Threshold (PWT), rats receiving intrathecal injection of vehicle (blue line), intrathecal injection of anti-LPA-mAb (10 ug/10µL) (red line), intravenous injection anti-LPA-mAb (10 mg/kg) (green line), the combination of both administrations intrathecal + intravenous of anti-LPA mAb (purple line) and intraperitoneal injection of gabapentin (50 mg/kg, aqua line). Data are expressed as mean, n = 8-9/group. (B) AUC Graph: Area under the 50% PWT curve (AUC) from 0 to 360 min for rats receiving intrathecal injection of vehicle (blue bar), intrathecal injection of anti-LPA-mAb (red bar), intravenous injection anti-LPA-mAb (green bar), the combination of intrathecal + intravenous of anti-LPA mAb (purple bar) and intraperitoneal injection of gabapentin (aqua bar). Data are expressed as mean±SEM, n = 8-9/group. \*\*p < 0.01 for anti-LPA i.t. 10 ug/10 µL + i.v. 10 mg/Kg v. Veh by one-way ANOVA with Bonferroni *post-hoc* analysis

**Figure 3** Concentration of Bioactive LPA in plasma

CHO-K1 EDG2 cells were plated as recommended and starved for 24 h. The plates were washed, and developed as recommended by the manufacture. Plasma samples from rats receiving intrathecal and intravenous injection of vehicle (Veh), or diabetic (D) rats randomized in the following groups: Veh + D, intrathecal and intravenous injection of vehicle, Low dose + D, intravenous injection anti-LPA-mAb (10 mg/kg) + intrathecal injection of anti-LPA-mAb (2 ug/10 µL) and High dose + D intravenous injection anti-LPA-mAb (10 mg/kg) + intrathecal injection of anti-LPA-mAb (10 ug/10 µL) were added to the plate and processed as described in methods. Data are expressed as mean  $\pm$  SEM, n = 10/group. \*\*p < 0.01 for Veh + D v. Veh and \*p < 0.05 for Low dose + D v. Veh by one-way ANOVA with Bonferroni post-hoc analysis

#### 4.4 CONCLUSIONS

The anti-LPA antibody, B3 increased the paw withdrawal threshold in diabetic rats to a level similar to that observed in non-diabetic rats, indicating a marked reduction in allodynia in rats with diabetes-induced neuropathic pain.

A single combined (i.t + i.v) dose of B3 was as effective as gabapentin in ameliorating established diabetic induced neuropathic pain.

The ability of B3 to mitigate pain responses from both routes of administration could suggest that LPA mediates both peripheral and central mechanisms of pain transmission.

There is an up regulation of circulating LPA levels in the plasma of diabetic animals and B3 treatment reduces bioactive LPA to concentrations comparable with control animals.

|                                                                                                          |               |
|----------------------------------------------------------------------------------------------------------|---------------|
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## **5 DATA ARCHIVE**

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## 6 APPENDICES

### Appendix 1 Raw Data

#### Tactile Allodynia: Prevention Study

|                                    |       |                                          |        |        |               |                                     |           |        |        |        |               |  |  |
|------------------------------------|-------|------------------------------------------|--------|--------|---------------|-------------------------------------|-----------|--------|--------|--------|---------------|--|--|
| Tactile Allodynia:                 |       | 4 week of diabetes/ 2 weeks of treatment |        |        |               |                                     |           |        |        |        |               |  |  |
|                                    |       | Paw withdrawal threshold                 |        |        |               |                                     |           |        |        |        |               |  |  |
| July 5, 2010                       |       |                                          |        |        |               |                                     |           |        |        |        |               |  |  |
| C+R+V: Control+Right+Vehicle       |       |                                          |        |        |               | D+R+V: Diabetic+Right+Vehicle       |           |        |        |        |               |  |  |
| C+L+V: Control+Left+Vehicle        |       |                                          |        |        |               | D+L+V: Diabetic+Left+Vehicle        |           |        |        |        |               |  |  |
|                                    |       | Both                                     |        |        |               |                                     |           | Both   |        |        |               |  |  |
| Veh+C                              |       | C+R+V                                    | C+L+V  | median | mean          |                                     | Veh+D     | D+R+V  | D+L+V  | median | mean          |  |  |
| 1.0                                | 15.00 | 11.70                                    | 13.35  | 13.35  |               | 5.0                                 | 6.66      | 3.33   | 4.99   | 4.99   |               |  |  |
| 1.1                                | 15.00 | 15.00                                    | 15.00  | 15.00  |               | 5.1                                 | 2.20      | 3.58   | 2.89   | 2.89   |               |  |  |
| 1.2                                | 11.70 | 15.00                                    | 13.35  | 13.35  |               | 6.0                                 | 4.98      | 2.81   | 3.89   | 3.89   |               |  |  |
| 2.0                                | 9.86  | 8.61                                     | 9.23   | 9.23   |               | 6.1                                 | 11.70     | 2.37   | 7.03   | 7.03   |               |  |  |
| 2.1                                | 15.00 | 15.00                                    | 15.00  | 15.00  |               | 7.1                                 | 4.47      | 6.42   | 5.44   | 5.44   |               |  |  |
| 3.0                                | 10.31 | 11.70                                    | 11.00  | 11.00  | July 12, 2010 | 12.0                                | 1.99      | 2.37   | 2.18   | 2.18   | July 12, 2010 |  |  |
| 3.1                                | 7.11  | 7.53                                     | 7.32   | 7.32   |               | 13.0                                | 4.40      | 4.25   | 4.32   | 4.32   |               |  |  |
| 3.2                                | 15.00 | 15.00                                    | 15.00  | 15.00  |               | 13.1                                | 6.64      | 6.58   | 6.61   | 6.61   |               |  |  |
| 4.0                                | 15.00 | 7.11                                     | 11.06  | 11.06  |               | 14.1                                | 4.25      | 4.25   | 4.25   | 4.25   |               |  |  |
| 4.1                                | 15.00 | 10.31                                    | 12.66  | 12.66  |               |                                     |           |        |        |        |               |  |  |
| median                             | 15.00 | 11.70                                    | 13.00  | 13.00  |               | median                              | 4.47      | 3.58   | 4.32   | 4.32   |               |  |  |
| mean                               | 12.90 | 11.70                                    | 12.30  | 12.30  |               | mean                                | 5.25      | 3.99   | 4.62   | 4.62   |               |  |  |
| sem                                | 0.93  | 1.02                                     | 0.83   | 0.83   |               | sem                                 | 0.97      | 0.53   | 0.50   | 0.50   |               |  |  |
|                                    |       |                                          |        |        |               |                                     |           |        |        |        |               |  |  |
| D+R+LD: Diabetic+Right+Low Dose    |       |                                          |        |        |               | D+R+HD: Diabetic+Right+Hight Dose   |           |        |        |        |               |  |  |
| D+L+LD: Diabetic+Left+Low Dose     |       |                                          |        |        |               | D+L+HD: Diabetic+Left+Hight Dose    |           |        |        |        |               |  |  |
|                                    |       | Both                                     |        |        |               |                                     |           | Both   |        |        |               |  |  |
| Low dose+D                         |       | D+R+LD                                   | D+L+LD | median | mean          |                                     | Hight dos | D+R+HD | D+L+HD | median | mean          |  |  |
| 8.0                                | 15.00 | 11.70                                    | 13.35  | 13.35  |               | 10.1                                | 5.37      | 3.67   | 4.52   | 4.52   |               |  |  |
| 9.0                                | 4.25  | 8.61                                     | 6.43   | 6.43   |               | 11.0                                | 6.42      | 6.58   | 6.50   | 6.50   |               |  |  |
| 9.1                                | 11.70 | 15.00                                    | 13.35  | 13.35  |               | 12.1                                | 15.00     | 15.00  | 15.00  | 15.00  |               |  |  |
| 10.0                               | 11.70 | 8.61                                     | 10.15  | 10.15  |               | 7.0                                 | 6.42      | 6.66   | 6.54   | 6.54   |               |  |  |
| 11.1                               | 15.00 | 15.00                                    | 15.00  | 15.00  |               | 17.0                                | 10.04     | 8.61   | 9.33   | 9.33   | July 12, 2010 |  |  |
| 14.0                               | 8.44  | 8.61                                     | 8.52   | 8.52   | July 12, 2010 | 17.1                                | 10.31     | 2.81   | 6.56   | 6.56   |               |  |  |
| 15.0                               | 15.00 | 13.96                                    | 14.48  | 14.48  |               | 18.0                                | 15.00     | 15.00  | 15.00  | 15.00  |               |  |  |
| 15.1                               | 6.58  | 6.58                                     | 6.58   | 6.58   |               | 18.1                                | 10.31     | 11.70  | 11.00  | 11.00  |               |  |  |
| 16.0                               | 8.61  | 7.94                                     | 8.28   | 8.28   |               | 19.0                                | 9.86      | 11.70  | 10.78  | 10.78  |               |  |  |
| 16.1                               | 15.00 | 11.70                                    | 13.35  | 13.35  |               | 19.1                                | 6.58      | 9.86   | 8.22   | 8.22   |               |  |  |
| median                             | 11.70 | 10.15                                    | 11.75  | 11.75  |               | median                              | 9.95      | 9.23   | 8.77   | 8.77   |               |  |  |
| mean                               | 11.13 | 10.77                                    | 10.95  | 10.95  |               | mean                                | 9.53      | 9.16   | 9.34   | 9.34   |               |  |  |
| sem                                | 1.26  | 0.98                                     | 1.05   | 1.05   |               | sem                                 | 1.09      | 1.36   | 1.14   | 1.14   |               |  |  |
| 2 ug/10ul, i.t.- Monday & Thursday |       |                                          |        |        |               | 10 ug/10ul, i.t.- Monday & Thursday |           |        |        |        |               |  |  |
| 10 mg/kg, i.v.- Tuesday & Friday   |       |                                          |        |        |               | 10 mg/kg, i.v.- Tuesday & Friday    |           |        |        |        |               |  |  |

## Tactile Allodynia: Intervention Study

|                                                                               |          |                                             |       |       |       |      |       |       |            |  |
|-------------------------------------------------------------------------------|----------|---------------------------------------------|-------|-------|-------|------|-------|-------|------------|--|
| Treatment: Diabetic rats were injected with Gabapentin 50 mg/kg (2 ml/Kg, ip) |          |                                             |       |       |       |      |       |       |            |  |
| July 19, 2010                                                                 |          | Gabapentin was dissolved in saline solution |       |       |       |      |       |       |            |  |
|                                                                               |          | Tactile Allodynia was measured every hour   |       |       |       |      |       |       |            |  |
|                                                                               |          |                                             |       |       |       |      |       |       |            |  |
|                                                                               |          |                                             |       |       |       |      |       |       |            |  |
| rat ID                                                                        | Baseline | 1h                                          | 2h    | 3h    | 4h    | 5h   | 6h    |       |            |  |
|                                                                               | 0        | 60                                          | 120   | 180   | 240   | 300  | 360   | AUC   | Cum. score |  |
| 22.0                                                                          | 1.99     | 7.94                                        | 6.58  | 6.58  | 6.58  | 7.11 | 5.37  | 38.47 | 42.15      |  |
| 23.0                                                                          | 3.67     | 11.70                                       | 11.70 | 8.44  | 10.31 | 5.56 | 6.64  | 52.85 | 58.00      |  |
| 23.1                                                                          | 3.67     | 11.70                                       | 11.70 | 11.70 | 11.70 | 7.11 | 11.70 | 61.58 | 69.26      |  |
| 24.0                                                                          | 1.85     | 6.58                                        | 5.04  | 8.44  | 4.47  | 3.72 | 1.56  | 29.94 | 31.64      |  |
| 25.0                                                                          | 1.99     | 1.56                                        | 2.81  | 4.47  | 2.20  | 1.56 | 3.33  | 15.24 | 17.90      |  |
| 25.1                                                                          | 4.40     | 6.66                                        | 11.70 | 6.66  | 4.25  | 5.04 | 2.37  | 37.68 | 41.06      |  |
| 1.0                                                                           | 3.67     | 4.47                                        | 11.70 | 8.61  | 6.58  | 1.99 | 2.81  | 36.57 | 39.81      |  |
| 1.1                                                                           | 2.37     | 6.58                                        | 11.70 | 11.70 | 11.70 | 8.61 | 5.56  | 54.23 | 58.19      |  |
| median                                                                        | 3.0      | 6.6                                         | 11.7  | 8.4   | 6.6   | 5.3  | 4.4   | 38.1  | 41.6       |  |
| mean                                                                          | 2.9      | 7.1                                         | 9.1   | 8.3   | 7.2   | 5.1  | 4.9   | 40.8  | 44.8       |  |
| sem                                                                           | 0.4      | 1.2                                         | 1.3   | 0.9   | 1.3   | 0.9  | 1.2   | 5.3   | 5.8        |  |

|                                          | 0     | 60   | 120  | 180  | 240  | 300  | 360  |
|------------------------------------------|-------|------|------|------|------|------|------|
| Veh                                      | 3.3   | 3.6  | 3.3  | 2.9  | 3.7  | 3.4  | 3.4  |
| SEM                                      | 0.4   | 0.5  | 0.3  | 0.3  | 0.7  | 0.6  | 0.6  |
| Anti-LPA mAb 10ug/10ul, it               | 3.7   | 4.1  | 6.8  | 5.6  | 5.6  | 4.5  | 4    |
| SEM                                      | 0.4   | 0.8  | 1.6  | 0.8  | 0.8  | 0.8  | 0.6  |
| Anti-LPA mAb 10mg/kg, iv                 | 3.2   | 4.4  | 5.5  | 5.7  | 3.6  | 4.8  | 4    |
| SEM                                      | 0.6   | 1.2  | 0.7  | 0.9  | 0.6  | 0.6  | 0.8  |
| Anti-LPA mAb 10ug/10ul, it + 10mg/kg, iv | 3.55  | 6.64 | 8.8  | 6.69 | 5.95 | 6.06 | 4.95 |
| SEM                                      | 0.51  | 1.4  | 1.49 | 0.99 | 1.16 | 1.03 | 1.13 |
| Gabapentin 50 mg/kg, ip                  | 2.9   | 7.1  | 9.1  | 8.3  | 7.2  | 5.1  | 4.9  |
| SEM                                      | 0.4   | 1.2  | 1.3  | 0.9  | 1.3  | 0.9  | 1.2  |
|                                          |       |      |      |      |      |      |      |
|                                          |       |      |      |      |      |      |      |
|                                          | AUC   | SEM  |      |      |      |      |      |
| Veh                                      | 20.20 | 2.06 |      |      |      |      |      |
| Anti-LPA mAb 10 ug, it                   | 30.46 | 3.16 |      |      |      |      |      |
| Anti-LPA mAb 10ug, it + 10mg/kg,         | 27.67 | 3.41 |      |      |      |      |      |
| Anti-LPA mAb 10ug, it + 10mg/kg,         | 39.85 | 6.03 |      |      |      |      |      |
| Gabapentin 50 mg/kg, ip                  | 40.82 | 5.28 |      |      |      |      |      |